

CPSC 481

Artificial Intelligence

Dr. Anand Panangadan
apanangadan@fullerton.edu

What we will cover today

- Bayesian networks in Python
 - <https://github.com/aimacode/aima-python>
- Code in folder “aima”
 - Though only need 3 files:
 - probability.py, utils.py, and agents.py

class BayesNet

- Stores the conditional probability tables (CPT) for each node
- Each node represents a Boolean variable (only T or F)
- When constructing a BN, add variables one at a time
 1. Provide the name of the variable
 2. Names of any parents
 3. The CPT for that node
 - CPTs are represented as Python dictionaries

('MaryCalls', 'Alarm', {T: 0.70, F: 0.01})

Inference algorithms

- Inference algorithms are functions
 1. `def enumeration_ask(X, e, bn)`
 2. `def elimination_ask(X, e, bn)`
 3. `def rejection_sampling(X, e, bn, N=10000)`
 4. `def gibbs_ask(X, e, bn, N=1000)`
- Each algorithm takes:
 - X: query variable
 - e: dictionary of evidence variable values
 - bn: the Bayesian network
- Each algorithm returns:
 - the probability distribution of X: $p(X \mid e)$

```
enumeration_ask('Burglary', dict(JohnCalls=T, MaryCalls=T), mybn)
```

Class work

- Set up a python environment (if you haven't)
 - Get started using Python on Windows for beginners
 - <https://docs.microsoft.com/en-us/windows/python/beginners>
 - Other environments
 - <https://www.python.org/downloads/>
 - or Google Colab
- Install **numpy** and **ipython** packages
- Download **aima** folder from
 - <https://github.com/aimacode/aima-python>

Create & execute “test_burglary.py”

```
from aima.probability import *

T, F = True, False

burglary = BayesNet([
    ('Burglary', '', 0.001),
    ('Earthquake', '', 0.002),
    ('Alarm', 'Burglary Earthquake',
     {(T, T): 0.95, (T, F): 0.94, (F, T): 0.29, (F, F): 0.001}),
    ('JohnCalls', 'Alarm', {T: 0.90, F: 0.05}),
    ('MaryCalls', 'Alarm', {T: 0.70, F: 0.01})
])

print(enumeration_ask('Burglary', dict(), burglary).show_approx())
print(enumeration_ask('Burglary', dict(JohnCalls=T, MaryCalls=T), burglary))
print(enumeration_ask('Burglary', dict(JohnCalls=T, MaryCalls=T), burglary)[T])
```

This code is also on Canvas

References

- AI textbook algorithms in Python
 - <https://github.com/aimacode/aima-python>
- Get started using Python on Windows for beginners
 - <https://docs.microsoft.com/en-us/windows/python/beginners>